Cover letter: B2 R2 submission

To: Editorial Office, Frontiers in Blockchain **From:** Zach Zukowski (Tokenization Systems; zach@tokenization.systems) **Re:** Tokenomics as Applied Political Philosophy: Governance Concentration Beyond Token Allocation (Manuscript ID 1853465; SSRN 6599278) **Title note (2026-05-22 update):** Title revised from the as-R2-submitted "Governance Concentration Beyond Token Allocation: An Institutional Design Analysis of DePIN and DeFi" to the current title per DEC-190 (Cycle 14 below). Empirical content and conclusions unchanged; framework-evidence integration elevated to the title surface. The short catalog form "Governance Concentration Beyond Token Allocation" (the subtitle of the new full title) preserves identity continuity with the prior title across cross-paper citations. **Date:** 2026-05-17 **Round:** R2 revision

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Dear Editor and Reviewers,

I am submitting the R2 revision of "Governance Concentration Beyond Token Allocation: An Institutional Design Analysis of DePIN and DeFi" addressing Reviewer 1's four R1-round comments and including substantial additional strengthening to the philosophical and methodological framework. This cover letter was authored 2026-05-17 to summarize the R1-reviewer-response cycle. Post-2026-05-17 internal hygiene cycles (deep-PCA-audit 2026-05-19; cycle 6 finalization 2026-05-21) further strengthened the manuscript findings beyond the R1-reviewer-flagged scope; those are documented in the "Post-2026-05-17 internal hygiene" section near the close of this letter, with per-surface numeric refreshes inline below where the R1-response narratives reference values that subsequently advanced.

Summary of R1 round-2 issues addressed

Reviewer 1's R1 round-2 review identified four classes of manuscript-vs-data-of-record drift. All four are addressed in this submission. The accompanying point-by-point response document (`2026-05-17_R2_responses_master.md`) details each fix at the level of granularity raised. In brief:

1. **Table 7 holding-HHI inconsistency**: Fully recomputed with post-exclusion holding HHIs consistent with Table 4. The R1-reviewer-flagged scope identified three protocols (Uniswap, Optimism, Lido) showing updated amplification ratios; two (Uniswap, Optimism) flip from delegation-dispersion to delegation-amplification under correct exclusion methodology, producing a substantive new finding (see below). The subsequent post-2026-05-17 deep-PCA-audit cycle (commit `3a8c3065`; 2026-05-19) extended this work by shifting Table 4 holding HHIs for nine protocols and expanding the Table 7 sample from N=8 to N=10 (added GMX and ENS via Tally additions per a 12-month governance-activity gate); see "Post-2026-05-17 internal hygiene" section below.

2. **Curve Top-10 column + CRV text disambiguation**: Top-10% corrected to post-exclusion value (34.4%). Body-text references to CRV 0.171 disambiguated: raw post-exclusion CRV HHI (0.017) is distinguished from veCRV-weighted voting concentration

(0.26, computed directly from Convex contract aggregation of top-50 historical lockers, a methodological strengthening over the R1 proxy estimate).

3. **Table 5 N=40 and TT-expanded reproducibility**: Hivemapper, io.net, and Aethir Gini values computed via Helius DAS API and Dune holder lists; all three now appear as full Table 4 rows. Table 5 HHI-Gini correlation now reports N=40 with complete data. TT-expanded subsidy N corrected from 22 (stale count) to 19 in the 2026-05-17 cycle; the post-2026-05-17 cycle 6 finalization workstream (cycle 12 net-flow operationalization 2026-05-19) further expanded the sample to N=20 with Pearson r=0.12, p=0.61, confirming the same qualitative null finding.
4. **Figure-vs-text statistical drift**: Manuscript surfaces now align on canonical values from a single regression dataset. Figure regeneration was in flight at 2026-05-17 cover letter authoring; subsequent cycles completed the regeneration (Fig 6 delegation amplification regenerated 2026-05-19 at commit `3a8c3065` with higher-precision raw_hhi inputs for figure-text alignment; Fig 9 LOO forest regenerated 2026-05-19 at commit `fb9ab9c6`). Regenerated figures are in the R2 submission package.

Substantive new finding emerging from R1 corrections

Applying the post-exclusion methodology consistently across Table 4 and Table 7 produces a **predominant-amplification finding** that evolved through three methodology cycles. The cycle-13-refreshed framing (2026-05-21) had eight of ten protocols amplifying with two structural exceptions (ENS at 0.45x; GMX at 0.87x); the 2026-05-22 multi-source PCA-symmetric extension (Cycle 15 below) expanded the Table 7 sample to N=13 (added Solana JUP delegated-vote, HNT VSR lockup-weighted, DRIFT VSR lockup-weighted) and the **current canonical framing is nine of thirteen protocols amplifying with four structural exceptions (ENS 0.45x mature-delegate-program; GMX 0.87x methodology-sensitive; HNT 0.35x-0.53x Solana VSR-lockup-weighted; JUP 0.057x most-extreme dispersion outlier with Solana delegated-vote plus broad airdrop user base mechanism)**. Amplification factors now range from 1.6x (DRIFT) to 9.9x (Compound) with mean approximately 4.9x for the nine amplifying protocols. JUP at 0.057x is 8x more extreme than the previous most-extreme outlier ENS, introducing a fundamentally new design pattern producing concentration order-of-magnitude below the cross-section median. Two protocols previously cited as delegation-dispersion counterexamples (Uniswap at 2.7x; Optimism at 3.6x) joined the amplification pattern under correct exclusion methodology in the cycle-13 reframe and remain in the amplifying set in Cycle 15. Curve and Balancer (vote-escrowed governance via veCRV and veBAL) show extreme amplification (15x and 21x respectively) reflecting lock-duration weighting and are discussed separately in Section 4.1 as a distinct ve-token class. The finding strengthens the paper's main thesis that institutional design at the operator and router level governs governance outcomes; magnitude-only variation within sector is a cleaner empirical signal than direction variation. Section 3.7 adds a PCA-symmetric robustness check extending across three governance-data sources (Tally all-delegates depth; Snapshot signer-side functional-PCA exclusion via new `exclusions_log_signer.csv` canonical store; Solana on-chain governance program parsing) confirming the finding holds across all Tally-sourced protocols even when foundation and aggregation-contract delegates are excluded.

Substantial R2 enhancements beyond reviewer-flagged issues

In addition to addressing all four R1 reviewer issues, the R2 cycle has substantively strengthened the philosophical-methodological framework:

- **Philosophical-credibility two-layer framing** (Section 2.10.1; Table 1): Each lens now carries an explicit operational mechanism (what is scored) and a philosophical goal (the normative warrant). This resolves the R1 Table 1 internal inconsistency where the Pettit row labeled "Contestation" while glossing "freedom from arbitrary power" (the definition of non-domination, not contestation). The two-layer structure prevents the common conflation of philosophical goal with institutional mechanism and is applied consistently across all five lenses.

- **Inter-lens relationships paragraph** (Section 2.9.6): New paragraph documenting that the five lenses are mutually reinforcing rather than orthogonal, providing the philosophical architecture for the Synergy Index aggregation choice.

- **Systematic empirical-philosophical mapping** (Section 4.1): For each major empirical finding, philosophical implications across applicable lenses are made explicit; the predominant delegation amplification finding (with ENS + GMX structural exceptions preserving the dispersion pattern as informative counterexamples; Section 3.5.1 + 3.5.1.1) registers across all five lenses simultaneously.

- **Comparative methodology** (Section 4.5): New paragraph positioning the five-lens framework against existing DAO assessment frameworks (Aragon DAO Health Index, Snapshot analytics, Boardroom, DAOstar standards) along three dimensions: scope, aggregation, comparability.

- **Falsifiable forward claims + pre-registration intent** (Section 4.9): Two specific predictions about subsequent cross-protocol governance trajectories that future panel-data work can test, with intent to pre-register hypothesis specifications.

- **Data availability and reproducibility statement** (Section 4.9): Explicit statement of replication-repository contents including the 40-protocol dataset, exclusions log, Phase 0 supplementary data (Hivemapper / io.net / Aethir holder distributions; veCRV cumulative deposits; Theil and Atkinson supplementary metrics).

- **Methodological depth additions** (Section 3.7): Theil (1967) and Atkinson (1970) inequality indices computed under PCA-symmetric methodology as a robustness check on HHI's value-validity; Shapley-Shubik (1954) and Banzhaf (1965) power indices acknowledged as voting-theory complements to HHI; Hirschman (1964) historical source for the HHI itself; multiple-comparisons correction note via Benjamini-Hochberg false-discovery-rate adjustment.

- **Operationalization-dependency caveat** (Section 4.8): Explicit acknowledgment that the framework's empirical conclusions depend on operationalization choices; alternative scoring criteria for the same lens could yield different lens scores.

- **21 new citations** across philosophy (Skinner, Cohen, Habermas, Berlin, Beitz, Pogge, Pettit 2012, Rawls 2001, Brennan & Buchanan, Hardin), mechanism design (Holmstrom, Tirole, North), voting theory (Shapley & Shubik, Banzhaf), DAO political theory (DuPont, Atzori, Schneider), and inequality measurement (Theil, Atkinson, Hirschman).

Submission package contents

- **B2_Frontiers_R2_clean.docx + B2_Frontiers_R2_clean.pdf**: Revised manuscript (clean version)
- **B2_Frontiers_R2_tracked_changes.docx + B2_Frontiers_R2_tracked_changes.pdf**: Revised manuscript with tracked changes from R1 baseline
- **2026-05-17_R2_responses_master.md (.docx + .pdf)**: Point-by-point response to Reviewer 1 comments
- **This cover letter**
- **Supplementary files (S1 through S17; expanded 2026-05-22 per word-cap trim cycle; see Cycle 14 below)**: Methodology and replication materials, including:
    - S1: Metric definitions including Synergy Index construction
    - S2: Scoring rubric templates with worked examples
    - S3: Per-protocol scoring sheets
    - S5: Replication pipeline specification
    - S6: Address-by-address exclusion methodology documentation
    - S7: Quarterly HHI panel for 14 governance tokens (8 quarters post-TGE)
    - S8: Token Terminal expansion subsidy sensitivity analysis
    - S9: Theil and Atkinson supplementary metrics (R2 addition)
    - S10 (NEW 2026-05-22): PCA-classification 5-spec robustness (per-specification narrative interpretation + alternative-classification rationale; main paper retains the 5-spec results table + interpretation summary)
    - S11 (NEW 2026-05-22): Banzhaf + Shapley-Shubik power-index calculations (full quorum-variation matrix + per-protocol Banzhaf-by-Shapley-Shubik divergence; main paper retains the Shapley-Shubik r = 0.999 vs voting-HHI summary + AAVE Banzhaf divergence)
    - S12 (NEW 2026-05-22): PCA-symmetric robustness check (per-protocol Compound Foundation + stkAAVE staking-aggregation case interpretations; main paper retains the predominant-amplification-holds summary)
    - S15 (NEW 2026-05-22): Optimism worked example full per-lens narrative (cited governance documentation; two-layer-framing caveat; main paper retains the condensed per-lens scoring + Synergy Index aggregation)
    - S16 (NEW 2026-05-22): Aethir / IoTeX / ENS sensitivity analyses (cycle 13 audit-trail; per-protocol PCA-exclusion edge-case detail; main paper retains the cascade-update summary)
    - S17 (NEW 2026-05-22): Falsifiable predictions detail (full operationalization per prediction + falsification thresholds + pre-registration commitment; main paper retains the 4-prediction brief list)

- Phase 0 R2 supplementary exhibits: holders_ATH_2026-05-17.csv, theil_atkinson_2026-05-17.csv, veCRV_voting_concentration_2026-05-17.csv, phase0_data_collection_results_2026-05-17.md

Methodology disclosure

The R2 cycle adopted a single-canonical-regression-dataset-of-record discipline to prevent recurrence of the manuscript-vs-data drift Reviewer 1 surfaced. The Section 2.10.3 methodology note documents this discipline. Snapshot-date discipline is acknowledged: the original Table 7 voting HHI values for the eight protocols sampled in the R1-reviewer-flagged scope use a March 2026 snapshot consistent with the rest of the dataset; the GMX and ENS additions from the post-2026-05-17 sample expansion use a May 2026 Tally snapshot (documented in Section 2.10.3 as a deliberate methodological exception for the sample-expansion protocols). The May 2026 Tally and Snapshot replication is consistent with delegate-pool drift (Compound active-delegate count and Snapshot voter count both shifted) but the rank ordering across protocols is the more durable inferential property than absolute magnitude.

Conflict of interest

The author served as Senior Investment Analyst at Borderless Capital, a cryptocurrency-focused investment firm, through February 2026 and retains no decision-making role, carried interest, or position-dependent compensation with the firm. The author may hold small personal positions in some protocols analyzed; positions were established outside the research period and are not position-dependent on Borderless Capital. A complete disclosure is provided in the Conflict of Interest statement accompanying this submission. No co-authors or other conflicts of interest.

Post-2026-05-17 internal hygiene

Subsequent to authoring this cover letter on 2026-05-17, two internal hygiene cycles ran against the manuscript without altering the R1-reviewer-flagged scope. Both strengthened (not weakened) the findings; the per-surface refreshes inline above reflect their effects.

Cycle 5 (2026-05-19; deep-PCA-audit + universal-audit cascade). The 126-address PCA audit (commit `19529d6a` workflow + `52146a1` replication) shifted Table 4 holding HHIs for nine protocols and expanded Table 7 from N=8 to N=10 (added GMX and ENS via Tally additions; 12-month governance-activity gate). Amplification range expanded from 1.9x-6.8x (mean 4.1x) to 1.2x-11.4x (mean 5.3x). The universal-amplification thesis reframes to near-universal: 9 of 10 protocols amplify; ENS is the single structural exception at 0.57x dispersion (mature delegate program where holders systematically delegate to a broader community-delegate set; Section 3.5.1 documents this as institutional-design-tractable rather than mechanism-inevitable). A subsequent universal-audit cycle (commit `a6d684c3`) refined ENS holding HHI from 0.071 to 0.049 via PCA exclusion of the ENS Cold Wallet (4.29M ENS, 5.2 percent of supply; verified on-chain as Safe Singleton 1.3.0 Smart Account; nick.eth creator), yielding the current 0.57x ratio. All R2 headline findings preserved across these cycles: allocation null (r=0.17, p=0.32, N=37); DePIN-DeFi sector contrast (Mann-Whitney p=0.020, Cohen’s d=0.94, LOO 30/30, permutation p=0.003); subsidy disconnect (Livepeer outlier); near-universal delegation amplification (with ENS exception). Figure 6 was regenerated with higher-precision raw_hhi inputs at this point.

Cycle 6 (2026-05-21; LFU 2300 absorption + cascade-audit). PAPER.md Section 3.4 + Section 3.6 stat surfaces (20 surfaces) cascaded to current canonical regression dataset values. One substantive direction-of-effect change emerged: the subsidy-excluding-Livepeer correlation reframes from “weakens substantially to r=0.45 / p=0.052 just above conventional 0.05 threshold” to “drops to clean null at r=0.03 / p=0.90 / N=24”; the Livepeer-driven-outlier finding strengthens. TT-expanded subsidy N expanded from N=19 to N=20 with corresponding Pearson r=0.12, p=0.61 confirming the same qualitative null finding. The sector contrast headline (Mann-Whitney p=0.020, Cohen’s d=0.94) was verified unchanged across both cycles. 16 calibrated-verb violations were cleared from PAPER.md prose during cycle 6 closeout follow-on; PAPER.md em-dash and calibrated-verb audits both return clean post-cycle.

Cycle 13 (2026-05-21; voting-HHI methodology refresh). Per a “Snapshot data may be corrupt + expand to at least N=1000” investigation, the voting-HHI methodology was refreshed comprehensively: Snapshot side pulled ALL proposals in the 12-month rolling window 2025-05-22 to 2026-05-22 (vs the pre-refresh 5-proposals-per-protocol sample), and Tally side expanded from top-100 to top-1000 delegates per protocol via paginated GraphQL queries. The refresh surfaced a substantive thesis shift: 8-of-10 protocols amplify (was 9-of-10); GMX flipped from amplification 1.2x (at Tally top-100) to dispersion 0.87x (at Tally top-1000); ENS dispersion deepened from 0.57x to 0.45x. The 1.2x GMX amplification finding in earlier cycles was a sampling-depth artifact, not a stable property of the underlying delegate-program design. Per-protocol ratio shifts: Compound 5.7x to 9.9x (largest amplification increase; Snapshot N expanded 114 to 138); Lido 11.4x to 6.3x (largest decrease; Snapshot N 262 to 333); Arbitrum 4.4x to 3.1x; Aave 5.9x to 4.8x; WeatherXM 3.3x to 3.8x; Uniswap 2.8x to 2.7x; DIMO 9.1x unchanged (genuine N=10). All R2 headline findings preserved: allocation null (r=0.05, p=0.76, N=37); DePIN-DeFi sector contrast (Mann-Whitney p=0.020, Cohen’s d=0.94); subsidy disconnect (Livepeer outlier); predominant delegation amplification (with ENS + GMX structural exceptions). NEW EC-2026-05-21-B2-Voting-HHI-Sampling-Methodology-Floor codified; F-B2-9 entry header reframed to predominant; NEW Section 3.5.1.1 GMX counterexample subsection added; Table 7 caption + Section 2.10.3 extended with refresh methodology documentation.

Method-of-record discipline: all three cycles followed the single-canonical-regression-dataset-of-record discipline established in the R2-2026-05-17 cycle (Section 2.10.3). The submission package included with this letter (B2_Frontiers_R2_clean.docx + .pdf; B2_Frontiers_R2_tracked_changes.docx + .pdf) reflects manuscript state post-cycle-13 voting-HHI methodology refresh.

Cycle 14 (2026-05-22; title rename + COI correction + word-cap trim cycle). Three post-R2 hygiene improvements landed in a single cycle 2026-05-22.

Title rename (DEC-190). The manuscript title was revised from “Governance Concentration Beyond Token Allocation: An Institutional Design Analysis of DePIN and DeFi” to “Tokenomics as Applied Political Philosophy: Governance Concentration Beyond Token Allocation” to better reflect the paper’s stated core thesis (Section 1.2). The empirical content, methodology, and conclusions are unchanged from the R2 resubmission shipped 2026-05-12. The five-lens philosophical framework (Kantian publicity, Rawlsian fairness, Republican non-domination, Ostromian polycentricity, Hayekian knowledge-use) operationalized against the 40-protocol cross-section remains the analytical core; the title rename elevates the framework-empirical integration to the title surface. SSRN v3 upload is in process (the SSRN v2 PDF at abstract ID 6599278 will be replaced with the post-rename PDF); Frontiers editor notification accompanying this update is documented at `handoff/author_instructions/B2_Frontiers_Editor_Title_Change_Notification_2026-05-22.md`.

COI correction. The Conflict of Interest statement in Section 2.10.8 was corrected to reflect that the author's tenure at Borderless Capital ended February 2026. The R2 manuscript carried stale "is employed by" framing; the corrected text reads: "The author served as Senior Investment Analyst at Borderless Capital, a cryptocurrency-focused investment firm, through February 2026 and retains no decision-making role, carried interest, or position-dependent compensation with the firm. The author may hold small personal positions in some of the protocols analyzed; positions were established outside the research period and are not position-dependent on Borderless Capital."

Word-cap trim cycle. In preparation for the Frontiers 12,000 word soft cap (the R2 manuscript was accepted for review at 17,702 words; the soft cap is treated as soft for substantive contributions), a targeted trim cycle reduced the body prose by ~1,381 words (~7.8% reduction) to ~16,321 words. The trim was structured as a hybrid supplementary migration: 6 new supplementary files (S10 through S17 enumerated in the Submission package contents section above) capture computational detail and per-protocol narrative depth that previously sat inline in the main paper, while inline summaries preserve all load-bearing inferential defenses and the framework-empirical bridges that the title rename specifically elevated. A post-trim audit cycle identified 5 substantive content losses where in-place compressions removed framework-empirical bridges rather than verbose methodology; all 5 were restored (cumulative restoration: +69 pandoc-equivalent words). Trim-effects assessment: directional findings preserved across the trim; statistical-significance reporting preserved verbatim; framework-evidence integration preserved per the title rename's editorial direction. Pre-trim binaries are preserved at `/tmp/B2_Pre_Word_Cap_Trim_2026-05-22_clean.{docx,pdf}` for rollback audit-trail.

Cycle 15 (2026-05-22; multi-source PCA-symmetric extension + max-effort deep-read audit + tracked-changes readability fix). A four-part cycle landed 2026-05-22 substantively strengthening reviewer-defensibility while preserving all R2 headline findings.

Multi-source PCA-symmetric extension (commits `1751fd5c` workflow + `d4366a1` replication). The §3.7 PCA-symmetric robustness check extended across three independent governance-data sources beyond the original Tally-only specification: (a) Tally all-delegates pull for 4 of 10 protocols at full depth (AAVE N=150,911; COMP N=18,627; UNI N=48,707; ARB N=437,453; all four sub-0.005 HHI shift vs cycle-13 top-1000 baseline, empirically validating the top-1000 methodology floor); (b) Snapshot 13-protocol voter-pool extension with cross-source label / Blockscout MCP / cross-protocol-overlap analysis (Tracks A through D); (c) Solana on-chain governance program parsing for JUP / HNT / DRIFT (JUP via Dune `jupiter_solana.govern_call_setvote` per-signer max-weight aggregation; HNT and DRIFT via Helius RPC parsing of SPL Governance VoteRecordV2 accounts). A new sibling canonical store `data/processed/exclusions_log_signer.csv` registers signer-side functional-PCA voters (Compound Foundation 0xb06df4dd…; Balancer DAO multisig 0xad9992f3…; Kevin Owocki / Gitcoin Maintainer 0x00de4b13…; WXM Hedgey-vested insider 0x96d8901a…; 4 confirmed entries after the DIMO Foundation candidate was

REFUTED via Polygon Blockscout MCP funding-source verification). A methodologically substantive finding: PCA-symmetric exclusion is NOT monotonic in the concentration-reduction direction; GTC HHI shifts from 0.1792 to 0.1948 upon signer-side PCA exclusion (smaller-denominator effect when excluded PCA holds dominant share). Three new EC entries codified: EC-2026-05-22-B2-WXM-Snapshot-Top1-Identification (hypothesis-revision audit trail); EC-2026-05-22-B2-HNT-Vote-Count-Proxy-vs-Lockup-Weight-Drift (methodology-floor drift; 5-7.5x shift between vote-count proxy and proper VSR lockup-weight); EC-2026-05-22-B2-DRIFT-Realms-Mint-Citation-Correction (surface-extraction-vs-on-chain-truth drift).

Table 7 N=10 to N=13 expansion + F-B2-9 reframe (commit `37635794`). Per the multi-source extension, three Solana-native protocols (JUP delegated-vote 0.057x dispersion; HNT VSR lockup-weighted 0.35x-0.53x sensitivity-range dispersion; DRIFT VSR lockup-weighted 1.6x mild amplification) were added to Table 7. Aave row updated from cycle-13 top-1000 0.062 to all-delegates 0.058 (4.4x ratio; methodology-floor empirical-validation tolerance under 0.005). F-B2-9 framing reframed from “8 of 10 amplify with 2 structural exceptions” to “9 of 13 amplify (range 1.6x DRIFT to 9.9x Compound; mean approximately 4.9x) with 4 structural exceptions (ENS 0.45x mature-delegate-program per Section 3.5.1; GMX 0.87x methodology-sensitive per Section 3.5.1.1; HNT 0.35x-0.53x Solana VSR-lockup-weighted per Section 3.5.4; JUP 0.057x most-extreme dispersion outlier with Solana delegated-vote + broad-airdrop-user-base mechanism per Section 3.5.4)”. Substantive thesis evolution: JUP at 0.057x is the new most-extreme dispersion outlier (8x more extreme than prior most-extreme ENS at 0.45x), introducing a fundamentally new design pattern (Solana delegated-vote + broad airdrop user base) producing concentration order-of-magnitude below the cross-section median. F-B2-9 predominant-amplification thesis PRESERVED at 9-of-13 majority. Three new PAPER.md subsections added: §3.5.4 within-Solana governance concentration heterogeneity; §3.5.4.1 Drift Foundation proposal-authorship versus vote-weight decoupling pattern; §3.5.5 cross-protocol institutional-governance-investor pattern (PGov + Tane + Arana ~17% combined Snapshot voting power; ~8% combined Tally voting power across ARB+COMP+UNI; candidate Fifth Concentration Axis alongside the four already in B2 scope). Figure 6 PNG regenerated with N=13 data via the canonical `tools/content-build/b2_figure6_regen.py` script; Figure 3 caption refreshed to current canonical (133 PCAs / 38 protocols post-cycle-4 audit; Aethir 0.095 / WeatherXM 0.148 / io.net 0.125 top DePIN cluster after IoTeX dropped from 0.189 to 0.081). 12+ prose surfaces cascaded with N=13 framing across abstract Finding 4, §1.4 fourth finding, §3.5 narrative, §3.7 amplification sub-text, §4.1 inter-lens implications, §4.6 summary of findings, §4.7 contributions, §5/6 governance design implications.

Max-effort deep-read stale-anchor audit (8 passes; ~80 anchors cleared). A systematic 8-pass deep-read audit cleared cumulative stale anchors arising from the multi-cycle revision arc: Pass 1 wording-variation residuals from the F-B2-9 reframe cascade (13 surfaces); Pass 2 Table 7 source-attribution completeness (Solana sources + cycle-13 voter counts; 1 footnote refresh); Pass 3 protocol-specific HHI cycle drift (7 surfaces; Curve 0.017 to 0.014 raw post-Convex-exclusion; Jupiter max-DeFi HHI 0.117 to 0.096; Hyperliquid minimum 0.004 to 0.005; Figure 3 caption pre-fix IoTeX 0.189 cluster reference); Pass 4 sequential-read in-context inconsistencies (13 surfaces; abstract Finding 4 enumeration; §1.4 amplifier list missing DRIFT; §3.3 DePIN mean 0.077 pre-IoTeX-fix to 0.071 post-fix; WeatherXM voting 0.486 to 0.556; “five DeFi protocols” to “seven”; Arbitrum amplification 4.4x typo to 3.1x; ENS top-1 19.9% pre-Cold-Wallet-fix to 16.8%); Pass 5 additional cross-checks (5 surfaces; Hivemapper 0.017 to 0.018;

Morpheus 0.031 to 0.046; AAVE delegation HHI 0.076 to 0.058; Figure 8 r=0.00 to r=-0.10; AAVE Shapley-Shubik methodology-mismatch acknowledgment); Pass 6 systematic Python-based Table 4 row vs sibling-clone replication CSV canonical (4 substantive row corrections; Aethir HHI 0.0868 to 0.0948 with updated Top-1 19.7% / Top-10 67.6%; LayerZero / Arbitrum / Optimism row refreshes); Pass 7 abstract-vs-Table 5 internal consistency (3 surfaces; Table 5 insider count fraction row rho=0.44 to 0.48 / p=0.005 to 0.003 / N=39 to 37 to match abstract canonical; LOO footnote 39/39 to 37/37); Pass 8 Table 4 N column convention per author directive (33 row updates: N now represents pre-exclusion sample size = 1,000 + number of PCAs; e.g., Optimism N = 1,007 reflects top-1,007 pre-exclusion sample with 7 PCAs excluded to yield 1,000 non-PCA holders for HHI computation). Plus Pass 8 extension: value-vs-convention methodology footnote acknowledging HHI / Top-N values reflect top-1000 pre-exclusion computation while N column reflects (1000+PCAs) convention; strict-convention recomputation would yield < 0.0001 HHI shift / < 0.1pp Top-N shift (immaterial for cross-protocol comparisons).

Substantive corrections caught in the audit (highlighted because they touch the manuscript's quantitative core): F-B2-14 (JUP outlier) "JUP-specific mechanisms rather than chain-level mechanisms explain the outlier" framing was internally inconsistent with the "low effective transaction cost" design implication (low tx cost IS a chain-level feature); refined to "JUP's design choices are protocol-design-level decisions ENABLED BY Solana's low-cost voting environment; whether they would replicate at scale on a higher-fee chain is untested because no Ethereum counterfactual exists" with a new EC-2026-05-22-B2-F-B2-14-Chain-vs-Protocol-Design-Dichotomy-Overclaim documenting the methodology lesson. F-B2-15 (within-Solana heterogeneity) DRIFT mis-classification corrected (was "delegated-vote with smaller professional user base"; correct is "VSR lockup-weighted vote design analogous to HNT" per DRIFT memo line 125); mechanism story reframed around VSR-vs-delegated as primary within-Solana driver per DRIFT memo line 130; sister EC-2026-05-22-B2-F-B2-15-DRIFT-VSR-Classification-And-Within-Solana-Mechanism-Refinement filed. The audit-pass discipline established by these corrections (verify just-shipped findings via Pattern 27 Mode A within-cycle) is itself an internal methodology contribution.

Tracked-changes readability fix (commit `859ebb64`). The previous tracked-changes generation wrapped EVERY text run in `w:ins` markers (8,462 markers) which made the tracked-changes view unreadable: every paragraph appeared edited, even those merely split from R1 mega-paragraphs during the jargon-refinement / accessibility cycles. A new diff-aware generator (`tools/content-build/regenerate_tracked_changes_v3.py`) performs paragraph-level token-set similarity matching between R1 baseline and R2 clean DOCX, wrapping `w:r` elements ONLY in paragraphs with substantive content changes (Jaccard >= 0.50 OR coverage >= 0.70 → unchanged; otherwise wrap). Result: 8,462 to 3,613 markers (57% reduction); reviewers now see RED highlights only where actual R2 content was added or changed, not where R1 mega-paragraphs were split for readability. The 192 marked paragraphs capture the substantive R2 changes (new §3.5.4 / 3.5.4.1 / 3.5.5 subsections; §3.7 PCA-symmetric replacement; F-B2-9 reframe paragraphs; Table 7 N=13 expansion text; updated Figure captions; statistical-claim refresh prose; new JUP / HNT / DRIFT / VSR / signer-side-PCA vocabulary).

Cycle 15 method-of-record discipline: single-canonical-regression-dataset-of-record discipline (Section 2.10.3) maintained throughout the multi-source extension; sibling clone `regression_data_april2026.csv` remains the holding-HHI canonical; new `exclusions_log_signer.csv` registered as sister canonical for signer-side voting-symmetric exclusion. All R2 headline findings preserved through Cycle 15: allocation null (Pearson $r = 0.05$, $p = 0.76$, $N = 37$); DePIN-DeFi sector contrast (Mann-Whitney $p = 0.020$, Cohen's $d = 0.94$; LOO 30/30); subsidy disconnect (Livepeer outlier; $r = 0.59$ inclusive / $r = 0.03$ exclusive); predominant delegation amplification (now 9 of 13 amplify; 4 structural exceptions: ENS + GMX + HNT + JUP); cross-protocol institutional concentration emerging as candidate Fifth Concentration Axis (per Section 3.5.5).

Cycle 16 (2026-05-22; revision-artifact cleanup + Frontiers 15-figure/table-limit compliance). A final manuscript-polish cycle landed 2026-05-22 immediately prior to this submission, removing revision-cycle annotations from the manuscript body and reducing the figure/table count to meet the Frontiers Original Research and Review Articles maximum of 15 figures/tables.

Revision-artifact cleanup (15+ surfaces). The manuscript body previously contained revision-cycle annotations carried forward from the multi-cycle revision process: parenthetical "(post-cycle-4 audit adding 0xfc78 Safe multisig)" in the Table 4 caption; "preserved as cycle-13 pre-deep-audit baseline" in the Figure 3 caption; "(post-refresh)" annotation on the §3.5.1.1 GMX header; "Lido's 6.3x amplification (post-refresh value per the 2026-05-21 voting-HHI methodology revision; pre-refresh 11.4x preserved as audit trail)" parenthetical; "in the R2 revision cycle" qualifier on the §3.5.4 Solana-protocols paragraph; Table 7 † footnote audit-trail "(cycle-13 refresh yields n = 138 … March 2026 baseline n = 114 preserved as audit-trail)"; Table 7 § footnote "(supersedes cycle-13 top-1000 value 0.062 with HHI shift -0.004 within the top-1000 methodology-floor empirical-validation tolerance)"; multi-paragraph §2.10.3 Snapshot date discipline revision-history narrative ("Compound grew from 114 in March 2026 to 138 by May 2026"); §3.5.1.1 GMX body "commit chain pre-2026-05-21" + "2026-05-21 voting-HHI methodology refresh" framing; §3.5 DeFi enumeration "at cycle-13 refresh" annotations; §3.7 PCA-symmetric paragraph "from the cycle-13 top-1000 sample baseline" annotations; §3.7 Shapley-Banzhaf "pre-2026-05-21 methodology refresh" parenthetical; §4.8 audit-cycle paragraph "The 2026-05-19 universal verification audit cycle surfaced three per-protocol PCA-exclusion sensitivity-check items…" multi-sentence revision-history paragraph; F-B2-9 internal identifier references (2 surfaces in §3.5.5 + §3.7); dated supplementary script file paths (7 surfaces; "b2/paper/supplements/synergy_index_full_sample_2026-05-19.py" etc. replaced with S-supplement prose references); dated price-snapshot references ("as of 2026-05-19" replaced with "May 2026 data-collection snapshot"). All revision-cycle annotations now condensed to current-state framing; audit-trail provenance preserved in this cover letter + the accompanying point-by-point response document + the linked supplementary materials. Verification post-cleanup: zero instances of cycle-13 / cycle-14 / cycle-15 / pre-refresh / post-refresh / post-cycle-4 / preserved-as-audit-trail / supersedes / multi-source-extension-cascade / pre-2026-05-21 / in-the-R2-revision-cycle / F-B2-N / EC-2026-NN / DEC-NNN patterns in the manuscript body.

Frontiers 15-figure/table-limit compliance. Pre-cleanup manuscript: 9 figures + 9 tables = 18 (over the Frontiers maximum of 15). Three items removed to bring the count to exactly 15: Figure 8 (HHI vs voter participation; $r = -0.10$ null; ancillary finding); Figure 9 (LOO forest plot for the DePIN-vs-DeFi sector contrast; visualizes the Cohen's d range and bootstrap 95% CI already discussed verbatim in §3.7 text); Table 9 (PCA-classification robustness; 5-spec exclusion sensitivity already documented in Supplementary File S10 with full per-specification narrative + alternative-classification rationale; inline content condensed to a single prose sentence citing the 5 specs by Mann-Whitney p + Cohen's d values). All three removed items meet four removability criteria simultaneously: (i) ancillary (not headline) finding; (ii) prose-vs-visual redundant (full content reproducible in 1-2 sentences); (iii) existing supplementary file alternative (S5 pipeline specification covers per-protocol participation + LOO iteration data; S10 covers PCA-classification robustness directly); (iv) low reviewer-defensibility cost (no headline finding lost). Post-cleanup manuscript: 7 figures (Figures 1-7) + 8 tables (Tables 1-8) = 15 (at limit). All headline figures preserved: Figure 5 lead-finding scatter (allocation null); Figure 4 sector boxplot (DePIN vs DeFi); Figure 6 delegation amplification (Table 7 visualization; N=13); Figure 7 subsidy outlier scatter (Livepeer-driven correlation); Figure 3 HHI bar 40 protocols (cross-section); Figures 1+2 framework diagrams.

Cycle 16 method-of-record discipline: revision-cycle annotations belong in submission-accompanying documents (this cover letter + the point-by-point response) and in the linked supplementary materials, not in the manuscript body. Reviewers reading the manuscript encounter current-state framing only; methodology evolution narratives are surfaced via this cover letter's cycle-by-cycle summary.

The submission package included with this letter (B2_Frontiers_R2_clean.docx + .pdf; B2_Frontiers_R2_tracked_changes.docx + .pdf) reflects manuscript state post-Cycle-16 with 15 figures/tables and revision artifacts removed.

Acknowledgment

We thank Reviewer 1 for the comprehensive cross-surface audit that identified the manuscript-vs-data-of-record drift and motivated the R2 reproduce-and-sync cycle, which in turn enabled the post-2026-05-17 deep-PCA-audit cycle to expand the Table 7 sample to N=10 and surface the ENS structural counterexample at 0.45x dispersion (Section 3.5.1). The predominant-amplification finding (with four structural exceptions ENS + GMX + HNT + JUP) documented in Section 3.5 is a direct product of Reviewer 1's R2 framing, extended via the post-2026-05-17 deep-PCA-audit cycle (commit `3a8c3065`), the 2026-05-21 voting-HHI methodology refresh (commit `99036422` per EC-2026-05-21-B2-Voting-HHI-Sampling-Methodology-Floor) which surfaced GMX as a 2nd structural-exception counterexample at expanded Tally top-1000 sampling depth, and the 2026-05-22 multi-source PCA-symmetric extension cycle (commit `1751fd5c`) which surfaced JUP as the most-extreme dispersion outlier (0.057x; Solana delegated-vote + broad airdrop user base mechanism) and HNT as a sensitivity-range dispersion case (0.35x-0.53x; Solana VSR lockup-weighted design) per the Section 3.5.4 within-Solana heterogeneity subsection. We thank Reviewer 2 for the endorsement of publication.

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